

Concise summary: the 5x3 matrix, the sectoral labour door, and the GAMMA menu

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Version 2 (September 2026) **Version 1** (September 2026)

One page of results, no derivation: the executed 5x3 matrix, the sectoral labour-market family that gives the flexible-wage rows a demand channel, and the seven fixed-wage (GAMMA) variants measured side by side. Everything below is a manifest field or a probe measurement; the sources are listed at the end, and the derivations live in `paper/equivalence.tex` (v3), `paper/framing_gamma.tex` and `docs/WORKPLAN_SENSITIVE_PRICES.md` (v7).

1 The 5x3 matrix, current results

Generation `matrix_5x3_v9` (33 cells, all gates pass). The fifteen matrix cells reproduce the `matrix_5x3_v6` values **bit-for-bit** (`max|Delta| = 0` on every metric and diagnostic), so the equivalence results of `paper/equivalence.tex` are invariant to the extension. Deviations from the no-programme baseline; the programme is 1.33 % of GDP.

Cell	Real GDP rel.	Consumption rel.	Employment	Deflator	max abs(p-1)
BF-F1	-0.000110	+0.000963	1.000000	1.006326	0.221442
BF-F2	-0.000162	-0.019800	1.000000	1.007181	0.279009
BF-F3	-0.000162	-0.019800	1.000000	1.007181	0.279009
ALPHA-F1	+0.000000	+0.001432	1.000000	1.000000	0.000000
ALPHA-F2	+0.000000	-0.018226	1.000000	1.000000	0.000000
ALPHA-F3	+0.000000	-0.018226	1.000000	1.000000	0.000000
BETA-F1	+0.000000	+0.001432	1.000000	1.000000	0.000000
BETA-F2	+0.000000	-0.018226	1.000000	1.000000	0.000000
BETA-F3	+0.000000	-0.018226	1.000000	1.000000	0.000000
GAMMA-F1	-0.000973	-0.000806	0.999027	1.000000	0.000000
GAMMA-F2	+0.001260	-0.015333	1.001260	1.000000	0.000000
GAMMA-F3	+0.017796	+0.022644	1.017796	1.000000	0.000000
DELTA-F1	-0.000973	-0.000806	0.999027	1.000000	0.000000
DELTA-F2	+0.001260	-0.015333	1.001260	1.000000	0.000000
DELTA-F3	+0.017796	+0.022644	1.017796	1.000000	0.000000

Three readings.

- **Only the BF row moves prices.** `max abs(p-1)` and the deflator are exactly zero/one in the twelve `eta = 1` cells of ALPHA, BETA, GAMMA and DELTA, and non-zero in BF (0.221-0.279; deflator 1.0063-1.0072). With one factor, constant returns and a demand-free price block, demand can reach prices only through an endogenous wage - which is why the repair belongs in the wage block.
- **The two equivalences are exact.** ALPHA = BETA (the elastic-supply row reproduces the full-employment row) and GAMMA = DELTA (the Leontief corner reproduces the fixed-wage row), both to all printed decimals in every financing column. Both are now *proved* ex ante, not merely observed (Propositions 1 and 2 of `paper/equivalence.tex`).
- **The financing spread is the live dimension in the fixed-wage row.** F3 is the Keynesian case (+1.78 % employment, +2.26 % consumption), F2 the tax-financed loss (-1.53 %), F1 near-neutral. In the mobile rows F2 and F3 are real-identical (`F_F3 = F_F2 - B_gov`).

2 The labour door: the sectoral family

N sectoral labour markets with supply elasticities $\eta_{s,i}$ (ADR-0022), whose **rigid corner is exactly the $\eta = 0$ endpoint** - the same equations, not a limit (Proposition 3 of `paper/equivalence.tex`; measured $\max|\Delta| = 0.00e+00$ warm-started, $< 1e-10$ cold, and bit-exact under a supply shock too). Executed in `matrix_5x3_v9`:

Sectoral cell	$\max \text{ abs}(p-1)$ F1 / F2 / F3	Employment (F2)	Consumption (F2)	Deflator (F2)	Wage max/min
uniform η_s = 0.25	0.133610 / 0.152897 / 0.152897	1.000864	-1.689%	1.004096	1.38
uniform η_s = 0.5	0.095839 / 0.105666 / 0.105666	1.001255	-1.571%	1.002872	1.26
uniform η_s = 1	0.061283 / 0.065422 / 0.065422	1.001627	-1.467%	1.001802	1.16
uniform η_s = 2	0.035629 / 0.037171 / 0.037171	1.001914	-1.391%	1.001033	1.09
rigid programme sectors, η_s = 0.5	0.215061 / 0.272033 / 0.272033	0.996621	-2.691%	1.006206	1.70
rigid largest half, η_s = 0.5	0.109583 / 0.123362 / 0.123362	1.000196	-1.905%	1.006294	1.28
BF ($\eta = 0$ endpoint)	0.221442 / 0.279009 / 0.279009	1.000000	-1.980%	1.007181	1.72
GAMMA (fixed wage, contrast)	0.000000 / 0.000000 / 0.000000	1.001260	-1.533%	1.000000	1.00

- **Demand sensitivity, defined strictly.** In every sectoral cell $\max \text{ abs}(p-1)$ differs between F1 and F2/F3 (F2 = F3 throughout), while all twelve $\eta = 1$ matrix cells read exactly zero. That difference - not the mere fact that prices move - is what the door delivers.
- **The elasticity splits the adjustment.** As η_s rises 0.25 to 2 the price response falls (0.1529 to 0.0372 at F2), employment rises (1.000864 to 1.001914) and the welfare cost falls (-1.69 % to -1.39 %).
- **Incidence dominates the level.** Making the seven programme sectors rigid more than doubles the price response (0.2720 against 0.1057) and *reverses* the employment effect (-0.34 % against +0.13 %).
- **The grouping rule is the open choice**, and it is now motivated rather than arbitrary: the programme's incidence is 63.4 % specialised construction works plus six material sectors, i.e. exactly the sectors where capacity pressure binds, so $\eta_{s,i} = 0$ there is a testable statement (`framing_gamma.tex` Section 4). The sweep gives the band: price response [0.097, 0.279], employment [-0.31 %, +0.16 %].
- **The level of η_s is unidentified under demand-only shocks** - the ladder is a sensitivity band. A supply shock separates the rungs (employment 1.000000 / 1.002187 / 1.008218 at $\eta_s = 0 / 0.5 / 2$), so the supply arm is the route to a number.

3 The GAMMA menu: seven fixed-wage variants

The second route to prices, and the key outcome of the latest round. The discriminator is the same: “sees demand” means $\max \text{ abs}(p-1)$ differs across the financing columns.

Variant	Financing	max abs(p-1)	Deflator	Employment	Consumption	Sees demand
1. baseline ($w = 1$)	F1	0.000000	1.000000	0.999027	-0.081%	no
	F2	0.000000	1.000000	1.001260	-1.533%	no
	F3	0.000000	1.000000	1.017796	+2.264%	no
2. DELTA (= baseline)	all	0.000000	1.000000	= baseline	= baseline	no
3. pinned wage structure (+10 % tilt)	all	0.057820	—	1.006865	—	no
4. capacity channel $\delta = +0.5$	F1	0.098457	1.003027	1.004167	+0.459%	yes
	F2	0.110761	1.005379	1.008489	-1.082%	yes
	F3	0.128463	1.021461	1.038814	+2.199%	yes
5. Kaldor- Verdoorn $\delta = -0.5$	F1	0.123462	1.003454	0.997837	-1.211%	yes
	F2	0.115629	1.001283	0.998527	-2.464%	yes
	F3	0.128014	0.985336	1.003370	+2.266%	yes
6. BF allocation rule ($\eta = 0$)	F1	0.000000	1.000000	1.000000	+0.043%	no
	F2	0.000000	1.000000	1.000000	-1.694%	no
	F3	0.000000	1.000000	1.000000	+0.000%	no
7. dual labour market (insiders rigid)	F1	0.024701	1.000438	0.989569	-1.338%	yes
	F2	0.027768	1.000492	0.991179	-2.881%	yes
	F3	0.030189	1.000648	1.005996	+0.689%	yes

- **Two variants see demand, through different margins.** The capacity channel through the *cost* (unit cost rises with own output); the dual labour market through *rationing* (insider prices 0.0278 against outsider 0.0015 at F2 - a 19x dualism signature, with identical wages in both segments).
- **The dual labour market is the only fixed-wage variant where the programme destroys employment** (-1.04 % / -0.88 % / +0.60 % across F1/F2/F3), because the bottleneck is transmitted economy-wide through intermediate costs. Quantity rationing is harsher than price-based rigidity (which reads -0.34 %).
- **The capacity channel's sign is the economics.** $\delta > 0$ is capacity pressure (F3 deflator 1.0215), $\delta < 0$ the Kaldor-Verdoorn case (F3 deflator 0.9853, prices falling with demand): a 3.6 pp deflator range from one parameter's sign.
- **The BF allocation rule is the double-rigidity corner.** Employment is a datum, prices are pinned, and under F3 household consumption is literally unchanged (+0.000 %). It is the benchmark the other variants are read against.
- **The pinned wage structure buys distribution, not demand** (prices move 0.058 identically in F1/F2/F3). It is the instrument for *who works where*.

4 The exact equivalences

Equivalence	Status	Source
ALPHA = BETA	proved ex ante; demand-only scope	Prop. 1, <code>equivalence.tex</code>
GAMMA = DELTA	definitional (Leontief corner)	Prop. 2, <code>equivalence.tex</code>

Equivalence	Status	Source
mobile F2 = F3	proved ($F_{F3} = F_{F2} - B_{gov}$)	<code>equivalence.tex</code> Section 5
sectoral corner = <code>eta</code> = 0 endpoint	proved, and shock-independent	Prop. 3, <code>equivalence.tex</code> v3
BF vs ALPHA	not a coincidence (ADR-0020 option C)	<code>equivalence.tex</code> classification

5 Open items

1. **The level of `eta_s`** (the supply-shock design; ADR-0018's schema item) - the one remaining batch that converts a band into a number.
2. **The grouping rule** - now motivated by capacity pressure, still to be preregistered with its testable evidence (vacancies, overtime, backlogs).
3. **The programme total — resolved (ADR-0024, 2026-09-20).** The impulse table is in **2023 prices** per Hornykewycz et al. (2025) (58 bn per year; first year 81 bn): 40.3 bn at 2019 prices *is* the horizon average (deflator 1.46; $58.89 / 1.46 = 40.338$ bn = the paper's exact 40.337 bn). $G0 = 40,300$ stays; shares and signs are unaffected, every level scales with it.
4. **The welfare metric** - dispersion-blind, and the kernel has no leisure term or income effect, so employment and wage dispersion are allocation facts, not welfare claims.
5. **The manuscript pass** and the `docs/DOCS_ASSESSMENT.md` block (still citing v3/v4 numbers).

6 Where the numbers come from

- Runs: `runs/matrix_5x3-v9-*` (33 cells; the matrix rows are bit-identical to `matrix_5x3-v6-*`). Flow tables: `paper/tables/matrix_5x3_v9_flows.md` and `matrix_5x3_v6_flows.md`.
- Probes: `probe13` / `probe14` (the sectoral closure and the S1-S5 slots), `probe15` (the supply arm), `probe16` (the rigidity band), `probe17` (the capacity channel in the fixed-wage closure), `probe18` (the BF allocation rule in GAMMA), `probe19` (the dual labour market).
- Decisions: ADR-0019 (external account), ADR-0020 (the `eta` = 0 endpoint), ADR-0021 (wage structure, proposed), ADR-0022 (sectoral labour markets, accepted), **ADR-0024 (programme total; price basis: 2023 prices per Hornykewycz et al. 2025; $G0 = 40,300$ at 2019 prices)**. Derivations: `paper/equivalence.tex` v3, `paper/framing_gamma.tex`. Steps and decision points: `docs/WORKPLAN_SENSITIVE_PRICES.md` v7.

7 Revision Log

- **Version 1** (September 2026)
- **Version 2 (September 2026)** — Open item 3 (programme total) resolved: the impulse table is in 2023 prices (Hornykewycz et al. 2025); 40.3 bn at 2019 prices is the horizon average (deflator 1.46, $58.89 / 1.46 = 40.338$ bn = 40.337 bn); $G0 = 40,300$ kept per ADR-0024 (accepted).